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Roll No

CS-3003-CBGS

B.E., III Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Digital Circuit and Design

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Simplify the following using Quine - McCluskey's method.
$$F(W, X, Y, Z) = \sum (0, 1, 2, 8, 10, 11, 14, 15)$$

b) State and prove De-Morgan's law.
2. a) Explain the minimization of switching function.
b) Prove sum of equation $Y = ABCD + ABC\bar{D}$ using Karnaugh Maps.
3. a) Design a full adder circuit with truth table and logic diagrams.
b) Why NAND and NOR gates are called universal gates? Implement AND, OR and NOT gate with universal gates.
4. a) How to design a combination circuit? Design a Full Adder circuit.
b) Draw and explain the working of monostable multivibrator.

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5.
 - a) What is counters? Differentiate Synchronous and Asynchronous counters.
 - b) What is shift registers? Explain.

6.
 - a) Implement a full adder circuit with a (3 to 8 lines) decoder and two OR gates.
 - b) Why Analog to Digital converters is needed? Explain any one Digital converters.

7.
 - a) Draw successive approximation analog to digital converter and explain its working in details.
 - b) With the help of circuit diagram explain the working of JK Flip Flop. Write its characteristics tables and equation.

8. Write a short notes on (any two):
 - a) PLA
 - b) Multiplexer
 - c) V-F converter
 - d) Encoder
